

BATTERY INPUT + PROTECTION

5V CONTROL BUCK (ESP32 + LOGIC ONLY)

WHEEL ENCODERS (3.3V)

BATTERY PI 5V / 5A BUCK – TPS54560B

RASPBERRY PI UART / CONTROL

SAFETY / REVIEW NOTES

3-DOF ARM SERVO CONTROL

LEFT MOTOR DRIVER

RIGHT MOTOR DRIVER

terminal, not 0.1in headers, for Pi power. Keep this 5A path short and wide.

reference design. Follow TI layout guidance; do not use this rail for motors or servos.

-PICO-KIT carrier uses the confirmed 17.78mm row spacing.

Encoder inputs include pull-ups, RC filtering, and Schmitt buffers.

- VNH5019 exposed pads need reflow, thermal vias, and large copper pours.
- Verify actual motor stall current before ordering production PCB.
- Pi power is NOT made by U1. Feed J8 from a separate regulated 5V/5A supply.
- Never power ESP32 from USB and EXT_5V simultaneously.
- External fuse and latching E-stop are mandatory.

Servo 1 = main swing; Servo 2 = elbow; Servo 3 = wrist. Pin order: GND / regulated 6V / PWM.

Pi power uses dedicated TPS54560B 5 V / 5 A buck
DRAFT: verify motor stall current and footprints before manufacture
3S LiPo; 2x GA25-370; provisional 5 A stall per motor

Harrison

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Title: Pickleball Robot Wheel Controller + Raspberry Pi Interface

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